

1 Absolute values and discrete valuations

1.1 Introduction

At its core, number theory starts with the ring $\mathbb{Z}$. By the fundamental theorem of arithmetic, every element of $\mathbb{Z}$ can be written uniquely as a product of primes (up to multiplication by a unit ± 1), so it is natural to focus on the prime elements of $\mathbb{Z}$. If p is a prime, the ideal $(p) := p\mathbb{Z}$ is a maximal ideal ($\mathbb{Z}$ has Krull dimension one), and the residue field $\mathbb{Z}/p\mathbb{Z}$ is the finite field $\mathbb{F}_p$ with p elements. The fraction field of $\mathbb{Z}$ is the field $\mathbb{Q}$ of rational numbers. The field $\mathbb{Q}$ and the finite fields $\mathbb{F}_p$ together make up the prime fields: every field k contains exactly one of them, according to its characteristic: k has characteristic zero if and only if it contains $\mathbb{Q}$, and k has characteristic p if and only if k contains $\mathbb{F}_p$.

One can also consider finite extensions of $\mathbb{Q}$, such as the field $\mathbb{Q}(i) := \mathbb{Q}[x]/(x^2+1)$. These are called *number fields*, and each can be constructed as the quotient of the polynomial ring $\mathbb{Q}[x]$ by one of its maximal ideals; the ring $\mathbb{Q}[x]$ is a principal ideal domain and its maximal ideals can all be written as (f) for some monic irreducible $f \in \mathbb{Z}[x]$.

Number fields are one of two types of *global fields* the others are *global function fields*. Let $\mathbb{F}_q$ denote the field with q elements, where q is any prime power. The polynomial ring $\mathbb{F}_q[t]$ has much in common with the integer ring $\mathbb{Z}$. Like $\mathbb{Z}$, it is a principal ideal domain of dimension one, and the residue fields $\mathbb{F}_q[t]/(f)$ one obtains by taking the quotient by a maximal ideal (f) , where $f \in \mathbb{F}_q[t]$ is any irreducible polynomial, are finite fields $\mathbb{F}_{q^d}$, where d is the degree of f . In contrast to the situation with $\mathbb{Z}$, the residue fields of $\mathbb{F}_q[t]$ all have the same characteristic as its fraction field $\mathbb{F}_q(t)$, which plays a role analogous to $\mathbb{Q}$. Global function fields are finite extensions of $\mathbb{F}_q(t)$.

Associated to each global field k is an infinite collection of *local fields* corresponding to the completions of k with respect to its absolute values; when $k = \mathbb{Q}$, these completions are the field of real numbers $\mathbb{R}$ and the p -adic fields $\mathbb{Q}_p$ (as you will prove on Problem Set 1).

The ring $\mathbb{Z}$ is a principal ideal domain (PID), as is $\mathbb{F}_q[t]$, and in such fields every nonzero prime ideal is maximal and thus has an associated *residue field*. For both $\mathbb{Z}$ and $\mathbb{F}_q[t]$ these residue fields are finite, but the characteristics of the residue fields of $\mathbb{Z}$ are all different (and distinct from the characteristic of its fraction field), while those of $\mathbb{F}_q[t]$ are all the same.

Remark 1.1. All rings have a multiplicative identity that is preserved by ring homomorphisms. Except where noted otherwise, the rings we shall consider are all commutative.

1.2 Absolute values

A reference for much of this material in this section is [3, Chapter 1].

Definition 1.2. An *absolute value* on a field k is a map $|\cdot|: k \rightarrow \mathbb{R}_{\geq 0}$ such that for all $x, y \in k$ the following hold:

1. $|x| = 0$ if and only if $x = 0$;
2. $|xy| = |x||y|$;
3. $|x + y| \leq |x| + |y|$.

If the stronger condition

4. $|x + y| \leq \max(|x|, |y|)$

also holds, then the absolute value is *nonarchimedean*; otherwise it is *archimedean*.

Example 1.3. The map $|\cdot|: k \rightarrow \mathbb{R}_{\geq 0}$ defined by

$$|x| = \begin{cases} 1 & \text{if } x \neq 0, \\ 0 & \text{if } x = 0, \end{cases}$$

is the *trivial absolute value* on k . It is nonarchimedean.

Lemma 1.4. An absolute value $|\cdot|$ on a field k is nonarchimedean if and only if

$$|\underbrace{1 + \cdots + 1}_n| \leq 1$$

for all $n \geq 1$.

Proof. See Problem Set 1. □

Corollary 1.5. In a field of positive characteristic every absolute value is nonarchimedean, and the only absolute value on a finite field is the trivial one.

Proof. Suppose k has positive characteristic p . Then every $n = 1 + \cdots + 1$ lies in the prime field $\mathbb{F}_p$ of k and satisfies $n^p = n$. This implies $|n|^p = |n|$, so $|n|$ is either 0 or 1 and k is nonarchimedean (by the lemma). If k is finite, say of cardinality q , then $x^q = x$ for all $x \in k$ and $|x|^q = |x|$ is 0 or 1, with the former occurring only for $x = 0$. □

Definition 1.6. Two absolute values $|\cdot|$ and $|\cdot|'$ on the same field k are *equivalent* if there exists an $\alpha \in \mathbb{R}_{>0}$ for which $|x|' = |x|^\alpha$ for all $x \in k$.

1.3 Absolute values on $\mathbb{Q}$

To avoid confusion we will denote the usual absolute value on $\mathbb{Q}$ (inherited from $\mathbb{R}$) by $|\cdot|_\infty$; it is an archimedean absolute value. But there are infinitely many others. Recall that any element of $\mathbb{Q}^\times$ may be written as $\pm \prod_q q^{e_q}$, where the product ranges over primes and the exponents $e_q \in \mathbb{Z}$ are uniquely determined (as is the sign).

Definition 1.7. For a prime p the *p-adic valuation* $v_p: \mathbb{Q} \rightarrow \mathbb{Z}$ is defined by

$$v_p\left(\pm \prod_q q^{e_q}\right) := e_p,$$

and we define $v_p(0) := \infty$. The *p-adic absolute value* on $\mathbb{Q}$ is defined by

$$|x|_p := p^{-v_p(x)},$$

where $|0|_p = p^{-\infty}$ is understood to be 0.

Theorem 1.8 (OSTROWSKI'S THEOREM). Every nontrivial absolute value on $\mathbb{Q}$ is equivalent to $|\cdot|_p$ for some $p \leq \infty$.

Proof. See Problem Set 1. □

Theorem 1.9 (PRODUCT FORMULA). For every $x \in \mathbb{Q}^\times$ we have

$$\prod_{p \leq \infty} |x|_p = 1.$$

Proof. See Problem Set 1. □

1.4 Discrete valuations

Definition 1.10. A *valuation* on a field k is a group homomorphism $k^\times \rightarrow \mathbb{R}$ such that for all $x, y \in k$ we have

$$v(x + y) \geq \min(v(x), v(y)).$$

We may extend v to a map $k \rightarrow \mathbb{R} \cup \{\infty\}$ by defining $v(0) := \infty$. For any $0 < c < 1$, defining $|x|_v := c^{v(x)}$ yields a nonarchimedean absolute value. The image of v in $\mathbb{R}$ is the *value group* of v . We say that v is a *discrete valuation* if its value group is equal to $\mathbb{Z}$ (every discrete subgroup of $\mathbb{R}$ is isomorphic to $\mathbb{Z}$, so we can always rescale a valuation with a discrete value group so that this holds). Given a field k with valuation v , the set

$$A := \{x \in k : v(x) \geq 0\},$$

is the *valuation ring* of k (with respect to v). A *discrete valuation ring* (DVR) is an integral domain that is the valuation ring of its fraction field with respect to a discrete valuation; such a ring A cannot be a field, since $v(\text{Frac } A) = \mathbb{Z} \neq \mathbb{Z}_{\geq 0} = v(A)$.

It is easy to verify that every valuation ring A is in fact a ring, and even an integral domain (if x and y are nonzero then $v(xy) = v(x) + v(y) \neq \infty$, so $xy \neq 0$), with k as its fraction field. Notice that for any $x \in k^\times$ we have $v(1/x) = v(1) - v(x) = -v(x)$, so at least one of x and $1/x$ has nonnegative valuation and lies in A . It follows that $x \in A$ is invertible (in A) if and only if $v(x) = 0$, hence the unit group of A is

$$A^\times = \{x \in k : v(x) = 0\},$$

We can partition the nonzero elements of k according to the sign of their valuation. Elements with valuation zero are units in A , elements with positive valuation are non-units in A , and elements with negative valuation do not lie in A , but their multiplicative inverses are non-units in A . This leads to a more general notion of a valuation ring.

Definition 1.11. A *valuation ring* is an integral domain A with fraction field k with the property that for every $x \in k$, either $x \in A$ or $x^{-1} \in A$.

Let us now suppose that the integral domain A is the valuation ring of its fraction field with respect to some discrete valuation v (which we shall see is uniquely determined). Any element $\pi \in A$ for which $v(\pi) = 1$ is called a *uniformizer*. Uniformizers exist, since $v(A) = \mathbb{Z}_{\geq 0}$. If we fix a uniformizer π , every $x \in k^\times$ can be written uniquely as

$$x = u\pi^n$$

where $n = v(x)$ and $u = x/\pi^n \in A^\times$ and uniquely determined. It follows that A is a unique factorization domain (UFD), and in fact A is a principal ideal domain (PID). Indeed, every nonzero ideal of A is equal to

$$(\pi^n) = \{a \in A : v(a) \geq n\},$$

for some integer $n \geq 0$. Moreover, the ideal (π^n) depends only on n , not the choice of uniformizer π : if π' is any other uniformizer its unique representation $\pi' = u\pi^1$ differs from π only by a unit. The ideals of A are thus totally ordered, and the ideal

$$\mathfrak{m} = (\pi) = \{a \in A : v(a) > 0\}$$

is the unique maximal ideal of A (and also the only nonzero prime ideal of A).

Definition 1.12. A *local ring* is a commutative ring with a unique maximal ideal.

Definition 1.13. The *residue field* of a local ring A with maximal ideal $\mathfrak{m}$ is the field $A/\mathfrak{m}$.

We can now see how to determine the valuation v corresponding to a discrete valuation ring A . Given a discrete valuation ring A with unique maximal ideal $\mathfrak{m}$, we may define $v: A \rightarrow \mathbb{Z}$ by letting $v(a)$ be the unique integer n for which $(a) = \mathfrak{m}^n$ and $v(0) := \infty$. Extending v to the fraction field k of A via $v(a/b) := v(a) - v(b)$ gives a discrete valuation v on k for which $A = \{x \in k : v(x) \geq 0\}$ is the corresponding valuation ring.

Notice that any discrete valuation v on k with A as its valuation ring must satisfy $v(\pi) = 1$ for some $\pi \in \mathfrak{m}$ (otherwise $v(k) \neq \mathbb{Z}$), and we then have $v(\pi) = 1$ if and only if $\mathfrak{m} = (\pi)$. Moreover, v must then coincide with the discrete valuation we just defined: for any DVR A , the discrete valuation on the fraction field of A that yields A as its valuation ring is uniquely determined. It follows that we could have defined a uniformizer to be any generator of the maximal ideal of A without reference to a valuation.

Example 1.14. For the p -adic valuation $v_p: \mathbb{Q} \rightarrow \mathbb{Z} \cup \{\infty\}$ we have the valuation ring

$$\mathbb{Z}_{(p)} := \left\{ \frac{a}{b} : a, b \in \mathbb{Z}, p \nmid b \right\},$$

with maximal ideal $\mathfrak{m} = (p)$; this is the *localization* of the ring $\mathbb{Z}$ at the prime ideal (p) . The residue field is $\mathbb{Z}_{(p)}/p\mathbb{Z}_{(p)} \simeq \mathbb{Z}/p\mathbb{Z} \simeq \mathbb{F}_p$.

Example 1.15. For any field k , the valuation $v: k((t)) \rightarrow \mathbb{Z} \cup \{\infty\}$ on the field of Laurent series over k defined by

$$v \left(\sum_{n \geq n_0} a_n t^n \right) = n_0,$$

where $a_{n_0} \neq 0$, has valuation ring $k[[t]]$, the power series ring over k . For $f \in k((t))^\times$, the valuation $v(f) \in \mathbb{Z}$ is the *order of vanishing* of f at zero. For every $\alpha \in k$ one can similarly define a valuation v_α on k as the order of vanishing of f at α by taking the Laurent series expansion of f about α .

1.5 Discrete Valuation Rings

Discrete valuation rings are in many respects the nicest rings that are not fields. In addition to being an integral domain, every discrete valuation ring A enjoys the following properties:

- *noetherian*: Every increasing sequence $I_1 \subseteq I_2 \subseteq \cdots$ of ideals eventually stabilizes; equivalently, every ideal is finitely generated.
- *principal ideal domain*: Every ideal is principal (generated by a single element).
- *local*: There is a unique maximal ideal $\mathfrak{m}$.
- *dimension one*: The (Krull) *dimension* of a ring R is the supremum of the lengths n of all chains of prime ideals $\mathfrak{p}_0 \subsetneq \mathfrak{p}_1 \subsetneq \cdots \subsetneq \mathfrak{p}_n$ (which need not be finite, in general). For DVRs, $(0) \subseteq \mathfrak{m}$ is the longest chain of prime ideals, with length 1.
- *regular*: The dimension of the $A/\mathfrak{m}$ -vector space $\mathfrak{m}/\mathfrak{m}^2$ is equal to the dimension of A . Non-local rings are regular if this holds for every localization at a prime ideal.
- *integrally closed* (or *normal*): Every element of the fraction field of A that is the root of a monic polynomial in $A[x]$ lies in A .

- *maximal*: There are no intermediate rings strictly between A and its fraction field.

Various combinations of these properties can be used to uniquely characterize discrete valuation rings (and hence give alternative definitions).

Theorem 1.16. *For an integral domain A , the following are equivalent:*

- A is a DVR.
- A is a noetherian valuation ring that is not a field.
- A is a local PID that is not a field.
- A is an integrally closed noetherian local ring of dimension one.
- A is a regular noetherian local ring of dimension one.
- A is a noetherian local ring whose maximal ideal is nonzero and principal.
- A is a maximal noetherian ring of dimension one.

Proof. See [1, §23] or [2, §9]. □

1.6 Integral extensions

Integrality plays a key role in number theory, so it is worth discussing it in more detail.

Definition 1.17. Given a ring extension $A \subseteq B$, an element $b \in B$ is *integral over A* if it is a root of a monic polynomial in $A[x]$. The ring B is *integral over A* if all its elements are.

Proposition 1.18. *Let $\alpha, \beta \in B$ be integral over $A \subseteq B$. Then $\alpha + \beta$ and $\alpha\beta$ are integral over A .*

Proof. Let $f \in A[x]$ and $g \in A[y]$ be such that $f(\alpha) = g(\beta) = 0$, where

$$\begin{aligned} f(x) &= a_0 + a_1x + \cdots + a_{m-1}x^{m-1} + x^m, \\ g(y) &= b_0 + b_1y + \cdots + b_{n-1}y^{n-1} + y^n. \end{aligned}$$

It suffices to consider the case

$$A = \mathbb{Z}[a_0, \dots, a_{m-1}, b_0, \dots, b_{n-1}], \quad \text{and} \quad B = \frac{A[x, y]}{(f(x), g(y))},$$

with α and β equal to the images of x and y in B , respectively, since given any $A' \subseteq B'$ we have homomorphisms $A \rightarrow A'$ defined by $a_i \rightarrow a_i$ and $b_i \rightarrow b_i$ and $B \rightarrow B'$ defined by $x \mapsto \alpha$ and $y \mapsto \beta$, and if $x + y, xy \in B$ are integral over A then $\alpha + \beta, \alpha\beta \in B'$ must be integral over A' .

Let k be the algebraic closure of the fraction field of A , and let $\alpha_1, \dots, \alpha_m$ be the roots of f in k and let $\beta_1, \dots, \beta_n$ be the roots of g in k . The polynomial

$$h(z) = \prod_{i,j} (z - (\alpha_i + \beta_j))$$

has coefficients that may be expressed as polynomials in the symmetric functions of the α_i and β_j , equivalently, the coefficients a_i and b_j of f and g , respectively. Thus $h \in A[z]$, and $h(x+y) = 0$, so $x+y$ is integral over A . Applying the same argument to $h(z) = \prod_{i,j} (z - \alpha_i\beta_j)$ shows that xy is also integral over A . □

Definition 1.19. Given a ring extension B/A , the ring $\tilde{A} = \{b \in B : b \text{ is integral over } A\}$ is the *integral closure of A in B* . When $\tilde{A} = A$ we say that A is *integrally closed in B* . For a domain A , its *integral closure* (or *normalization*) is its integral closure in its fraction field, and A is *integrally closed* (or *normal*) if it is integrally closed in its fraction field.

Proposition 1.20. *If $C/B/A$ is a tower of ring extensions in which B is integral over A and C is integral over B then C is integral over A .*

Proof. See [1, Thm. 10.27] or [2, Cor. 5.4]. $\square$

Corollary 1.21. *If B/A is a ring extension, then the integral closure of A in B is integrally closed in B .*

Proof. Let A' be the integral closure of A in B and let A'' be the integral closure of A' in B . Proposition 1.20 implies that A'' is integral over A . Every element of $B \supset A''$ that is integral over A lies in A' (by definition), so $A' \subseteq A'' \subseteq A'$. Thus A' is equal to its integral closure (hence integrally closed) in B . $\square$

Proposition 1.22. *The ring $\mathbb{Z}$ is integrally closed.*

Proof. We apply the rational root test: suppose $r/s \in \mathbb{Q}$ is integral over $\mathbb{Z}$, where r and s are coprime integers. Then

$$\left(\frac{r}{s}\right)^n + a_{n-1}\left(\frac{r}{s}\right)^{n-1} + \cdots + a_1\left(\frac{r}{s}\right) + a_0 = 0$$

for some $a_0, \dots, a_{n-1} \in \mathbb{Z}$. Clearing denominators yields

$$r^n + a_{n-1}sr^{n-1} + \cdots + a_1s^{n-1}r + a_0s^n = 0,$$

thus $r^n = -s(a_{n-1}r^{n-1} + \cdots + a_1s^{n-2}r + a_0s^{n-1})$ is a multiple of s . But r and s are coprime, so $s = \pm 1$ and therefore $r/s \in \mathbb{Z}$. $\square$

Corollary 1.23. *Every unique factorization domain is integrally closed. In particular, every PID is integrally closed.*

Proof. The proof of Proposition 1.22 works for any UFD. $\square$

Example 1.24. The ring $\mathbb{Z}[\sqrt{5}]$ is not a UFD (nor a PID) because it is not integrally closed: consider $\phi = (1 + \sqrt{5})/2 \in \text{Frac } \mathbb{Z}[\sqrt{5}]$, which is integral over $\mathbb{Z}$ (and hence over $\mathbb{Z}[\sqrt{5}]$), since $\phi^2 - \phi - 1 = 0$. But $\phi \notin \mathbb{Z}[\sqrt{5}]$, so $\mathbb{Z}[\sqrt{5}]$ is not integrally closed.

It follows that every discrete valuation ring is integrally closed. In fact, more is true.

Proposition 1.25. *Every valuation ring is integrally closed.*

Proof. Let A be a valuation ring with fraction field k and let $\alpha \in k$ be integral over A . Then

$$\alpha^n + a_{n-1}\alpha^{n-1} + a_{n-2}\alpha^{n-2} + \cdots + a_1\alpha + a_0 = 0$$

for some $a_0, a_1, \dots, a_{n-1} \in A$. Suppose $\alpha \notin A$. Then $\alpha^{-1} \in A$, since A is a valuation ring. Multiplying the equation above by $\alpha^{-(n-1)} \in A$ and moving all but the first term on the LHS to the RHS yields

$$\alpha = -a_{n-1} - a_{n-1}\alpha^{-1} - \cdots - a_1\alpha^{2-n} - a_0\alpha^{1-n} \in A,$$

contradicting our assumption that $\alpha \notin A$. It follows that A is integrally closed. $\square$

Definition 1.26. A *number field* K is a finite extension of $\mathbb{Q}$. The *ring of integers* $\mathcal{O}_K$ is the integral closure of $\mathbb{Z}$ in K .

Remark 1.27. The notation $\mathbb{Z}_K$ is also sometimes used to denote the ring of integers of K . The symbol $\mathcal{O}$ emphasizes the fact that $\mathcal{O}_K$ is an *order* in K ; in any $\mathbb{Q}$ -algebra K of finite dimension r , an order is a subring of K that is also a free $\mathbb{Z}$ -module of rank r , equivalently, a $\mathbb{Z}$ -lattice in K that is also a ring. In fact, $\mathcal{O}_K$ is the *maximal order* of K : it contains every order in K .

Proposition 1.28. Let A be an integrally closed domain with fraction field K . Let α be an element of a finite extension L/K , and let $f \in K[x]$ be its minimal polynomial over K . Then α is integral over A if and only if $f \in A[x]$.

Proof. The reverse implication is immediate: if $f \in A[x]$ then certainly α is integral over A . For the forward implication, suppose α is integral over A and let $g \in A[x]$ be a monic polynomial for which $g(\alpha) = 0$. In $\overline{K}[x]$ we may factor $f(x)$ as

$$f(x) = \prod_i (x - \alpha_i).$$

For each α_i we have a field embedding $K(\alpha) \rightarrow \overline{K}$ that sends α to α_i and fixes K . As elements of $\overline{K}$ we have $g(\alpha_i) = 0$ (since $f(\alpha_i) = 0$ and f must divide g), so each $\alpha_i \in \overline{K}$ is integral over A and lies in the integral closure $\tilde{A}$ of A in $\overline{K}$. Each coefficient of $f \in K[x]$ can be expressed as a sum of products of the α_i , and is therefore an element of the ring $\tilde{A}$ that also lies in K . But $A = \tilde{A} \cap K$, since A is integrally closed in its fraction field K . $\square$

Example 1.29. We saw in Example 1.24 that $(1 + \sqrt{5})/2$ is integral over $\mathbb{Z}$. Now consider $\alpha = (1 + \sqrt{7})/2$. Its minimal polynomial $x^2 - x - 3/2 \notin \mathbb{Z}[x]$, so α is not integral over $\mathbb{Z}$.

References

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- [3] Jean-Pierre Serre, *Local fields*, Springer, 1979.